THE INTERNAL HOM OF A STRICTLY PERFECT COMPLEX WITH ANY COMPLEX IS REPRESENTED BY THE TERMWISE INTERNAL HOM COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the choice of the K-injective resolution, the product complex representing the derived internal hom, the identification of the sheafified H^0 through the local comparison lemma, and the extension to all degrees giving the quasi-isomorphism.

Editorial note. The label lemma-Rhom-strictly-perfect names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Strictly perfect complexes and the internal hom. Same chapter, the definition of a strictly perfect complex and the section constructing the internal hom complex with its differential. Supplies the hypotheses on E^bullet and the differential of the representing complex in the statement.
- K-injective resolutions and the product complex representing R SheafHom. Same chapter, the construction representing R SheafHom by the product complex of internal homs into a K-injective resolution. Supplies the complex (H')^bullet used in the proof and its quasi-isomorphism to R SheafHom.
- Local comparison of Hom sets for strictly perfect complexes. Same chapter, the lemma identifying the sheafification of the local Hom functor with the sheafification of the derived Hom for strictly perfect complexes. Supplies that the two complexes have the same sheafified cohomology, which is what makes them quasi-isomorphic.

Source reference. Original reference 08C9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11881–11881. Prerequisite pointer invoked at proof

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line(s) 11950; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 017G: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8836–8836. Prerequisite pointer invoked at proof line(s) 11923; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{E}^\bullet$, $\mathcal{F}^\bullet$ be complexes of $\mathcal{O}$-modules with $\mathcal{E}^\bullet$ strictly perfect. Then the internal hom $R\mathcal{H}om(\mathcal{E}^\bullet, \mathcal{F}^\bullet)$ is represented by the complex $\mathcal{H}^\bullet$ with terms*

$$\mathcal{H}^n = \bigoplus\nolimits_{n=p+q} \mathcal{H}om_{\mathcal{O}}(\mathcal{E}^{-q}, \mathcal{F}^p)$$

and differential as described in Section 017G.

Proof. Choose a quasi-isomorphism $\mathcal{F}^\bullet \to \mathcal{I}^\bullet$ into a K-injective complex. Let $(\mathcal{H}')^\bullet$ be the complex with terms

$$(\mathcal{H}')^n = \prod\nolimits_{n=p+q} \mathcal{H}om_{\mathcal{O}}(\mathcal{L}^{-q}, \mathcal{I}^p)$$

which represents $R\mathcal{H}om(\mathcal{E}^\bullet, \mathcal{F}^\bullet)$ by the construction in Section 017G. It suffices to show that the map

$$\mathcal{H}^\bullet \longrightarrow (\mathcal{H}')^\bullet$$

is a quasi-isomorphism. Given an object U of $\mathcal{C}$ we have by inspection

$$H^0(\mathcal{H}^\bullet(U)) = \mathrm{Hom}_{K(\mathcal{O}_U)}(\mathcal{E}^\bullet|_U, \mathcal{I}^\bullet|_U) \to H^0((\mathcal{H}')^\bullet(U)) = \mathrm{Hom}_{D(\mathcal{O}_U)}(\mathcal{E}^\bullet|_U, \mathcal{I}^\bullet|_U)$$

By Lemma 08C9 the sheafification of $U \mapsto H^0(\mathcal{H}^\bullet(U))$ is equal to the sheafification of $U \mapsto H^0((\mathcal{H}')^\bullet(U))$. A similar argument can be given for the other cohomology sheaves. Thus $\mathcal{H}^\bullet$ is quasi-isomorphic to $(\mathcal{H}')^\bullet$ which proves the lemma. □